

Project 19

Apparent Barriers in iiwa 7 Redundancy Charts

Format: Group of two students; simulation-based project.

Prerequisites: Redundant inverse kinematics, self-motion coordinates, reduced and full Jacobians, and numerical continuation.

Motivation

Fixing a redundant joint converts a 7R IK problem into a square subproblem, but that subproblem can become singular while the full robot remains regular. A planner may then report an apparent barrier caused by its chosen coordinate. A local change of coordinate can reveal whether the underlying robot motion continues.

Task

Overall objectives. Analyze a supplied iiwa path with a fixed- q_3 chart failure and determine whether a second redundancy coordinate can continue the same full-pose IK family without violating robot constraints.

- **Reproduce the supplied case.** Use the instructor-provided short pose path, initial IK, and suspected chart-fold interval. Validate the iiwa URDF/D-H model and reproduce the failure of fixed- q_3 continuation with documented residuals, limits, and solver tolerances.
- **Separate the two regularity questions.** Compare $\sigma_{\min}(J)$ for the scaled full 6×7 Jacobian with $\det \bar{J}_3$, where $\bar{J}_3$ removes column 3. Explain how a singular reduced matrix can reflect tangency of the self-motion family to the chosen coordinate rather than rank loss of the full robot.
- **Construct a second local chart.** Select one alternative joint q_j for which the corresponding six-column Jacobian is well conditioned near the supplied interval. Hold q_j fixed in a local numerical corrector and continue the same recovered configuration family. Use only this one alternative chart and a bounded overlap region.
- **Classify and reconcile the charts.** Generate path-sample versus redundant-angle panels for q_3 and for q_j . Preserve the supplied fixed- q_3 labels and assign traceable local continuation identities in the second chart. Match overlapping solutions by full joint configuration and pose closure, not by equal labels or determinant signs alone.
- **Validate the crossing or obstruction.** Attempt one trajectory using the chart overlap and compare with the original method. Check intermediate poses, full-Jacobian regularity, limits, and continuity. Mark which chart barrier was removed and which failures persist; failure of two charts alone does not prove robot infeasibility.

Specifics and scope

The instructor will supply the path, starting solution, and a known candidate fold interval. Limit the study to two coordinates, one overlap, and one local chart transition, with roughly 80 path samples and 100 angle samples per chart initially. Reuse the fixed- q_3 analytical IK; a local numerical corrector is sufficient for the alternative coordinate.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand coordinate-induced IK ambiguity and distinguish it from a physical singularity or constraint. Deliver two classified feasibility charts, overlap matches, full/reduced regularity plots, and a validated continuation or an explicit account of the unresolved obstruction.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 robot description (URDF) and visualization meshes (STL) will be provided to students, together with a mathematical continuation model with unrestricted joints and an analytical fixed- q_3 IK model. Reuse the supplied IK rather than deriving a general 7R solver. Apply the physical joint limits from the robot description separately when using the unrestricted model. Check D–H parameters, joint axes, and the fixed tool transform against the URDF, and validate IK outputs by independent forward kinematics.

The special short pose path, starting IK, and candidate chart-fold interval will be supplied. Use this case to initialize both continuation methods; an arbitrary reference path is not guaranteed to contain the required event. For pose data represented by quaternions, use the documented scalar-first columns $\mathbf{qw}, \mathbf{qx}, \mathbf{qy}, \mathbf{qz}$, normalize them, and make their signs consistent before interpolation.

Implement the alternative-coordinate numerical corrector and overlap matching around the supplied fixed- q_3 IK. Compare the reduced and full Jacobians, check intermediate pose residuals and physical limits, and visualize the recovered motion with the supplied meshes.